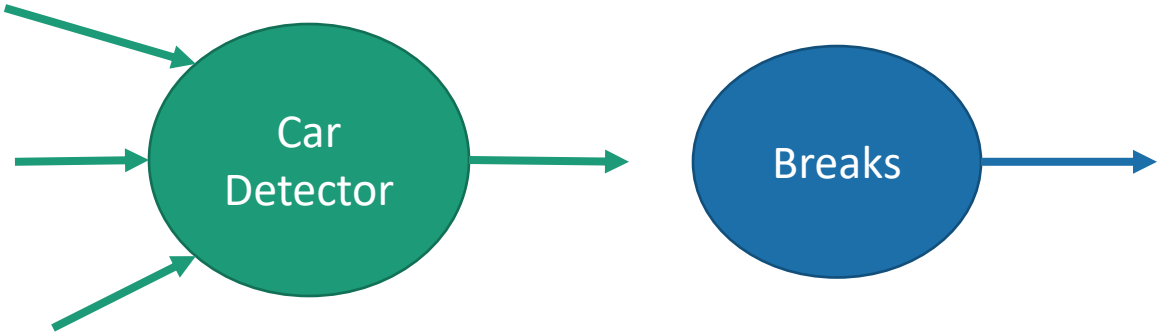
Inhibition and excitation

- Grace Lindsay – Chapter 5

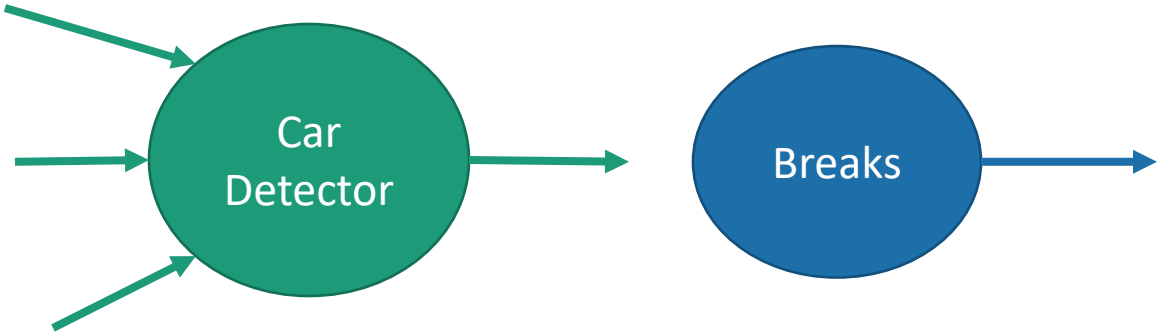
Noise

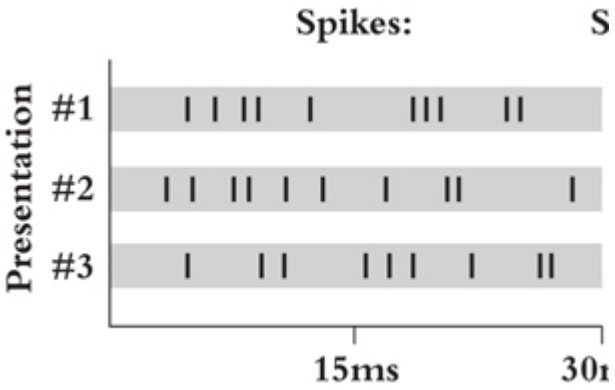




Noise







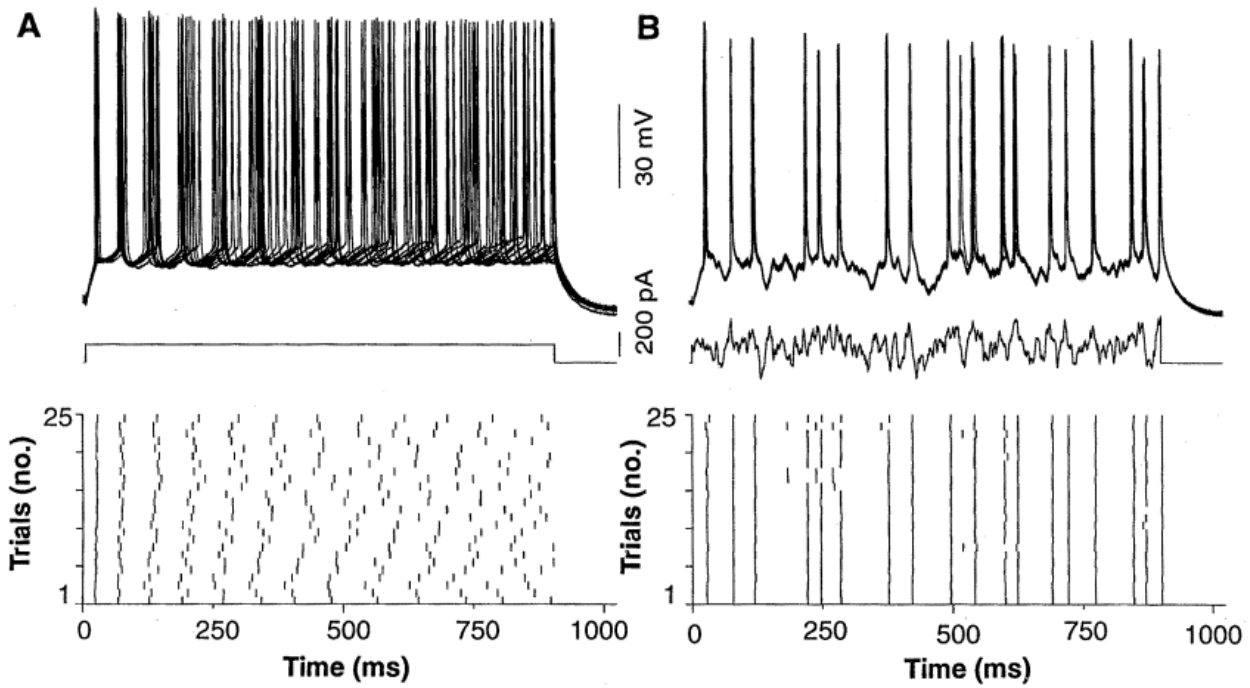


Fig. 1. Reliability of firing patterns of cortical neurons evoked by constant and fluctuating current. (**A**) In this example, a superthreshold dc current pulse (150 pA, 900 ms; middle) evoked trains of action potentials (approximately 14 Hz) in a regular-firing layer-5 neuron. Responses are shown superimposed (first 10 trials, top) and as a raster plot of spike times over spike times (25 consecutive trials, bottom). (**B**) The same cell as in (A) was again stimulated repeatedly, but this time with a fluctuating stimulus [Gaussian white noise, μ_s = 150 pA, σ_s = 100 pA, τ_s = 3 ms; see (*14*)].

Reliability of Spike Timing in Neocortical Neurons Zachary F. Mainen* and Terrence J. Sejnowski

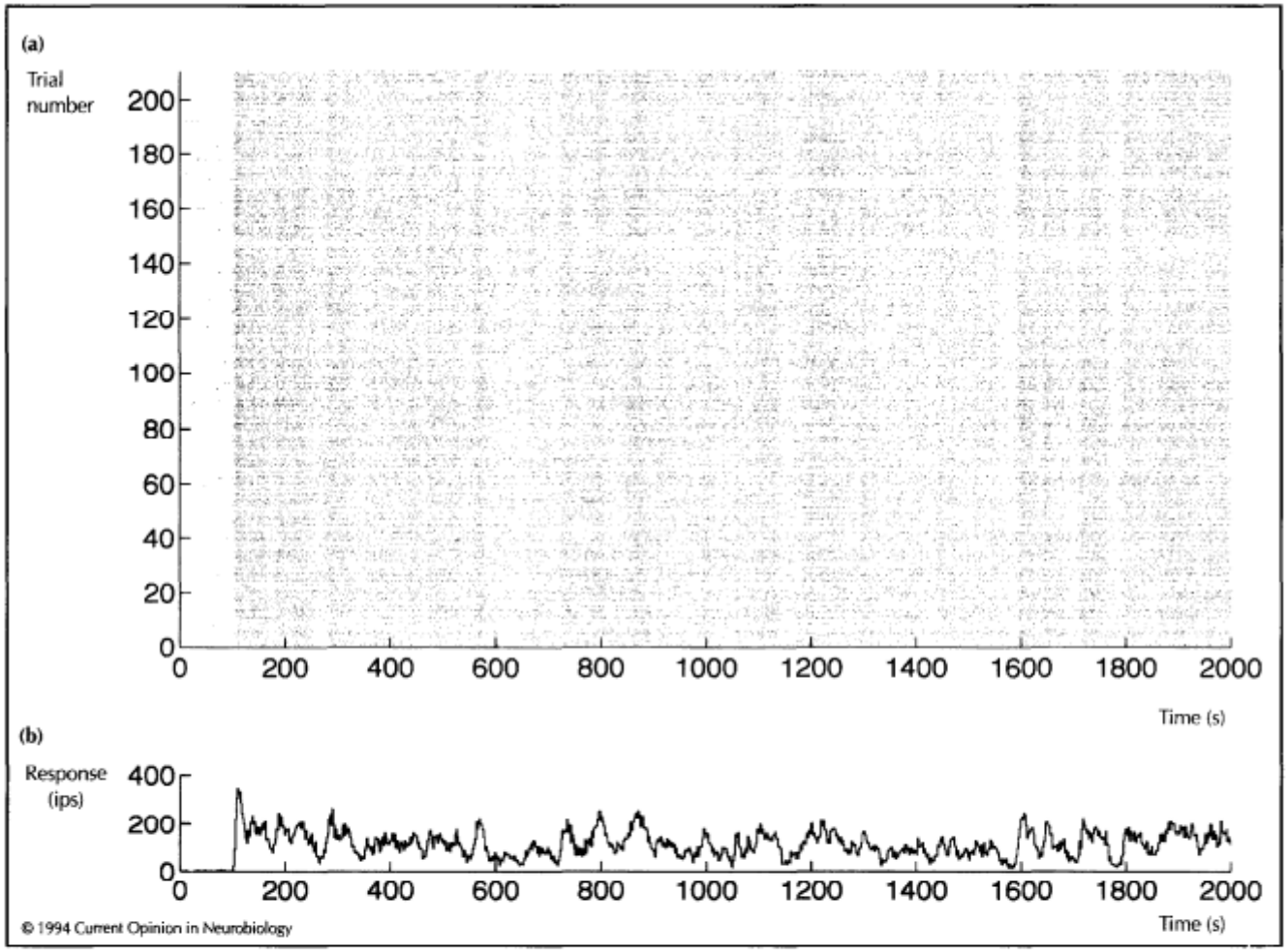
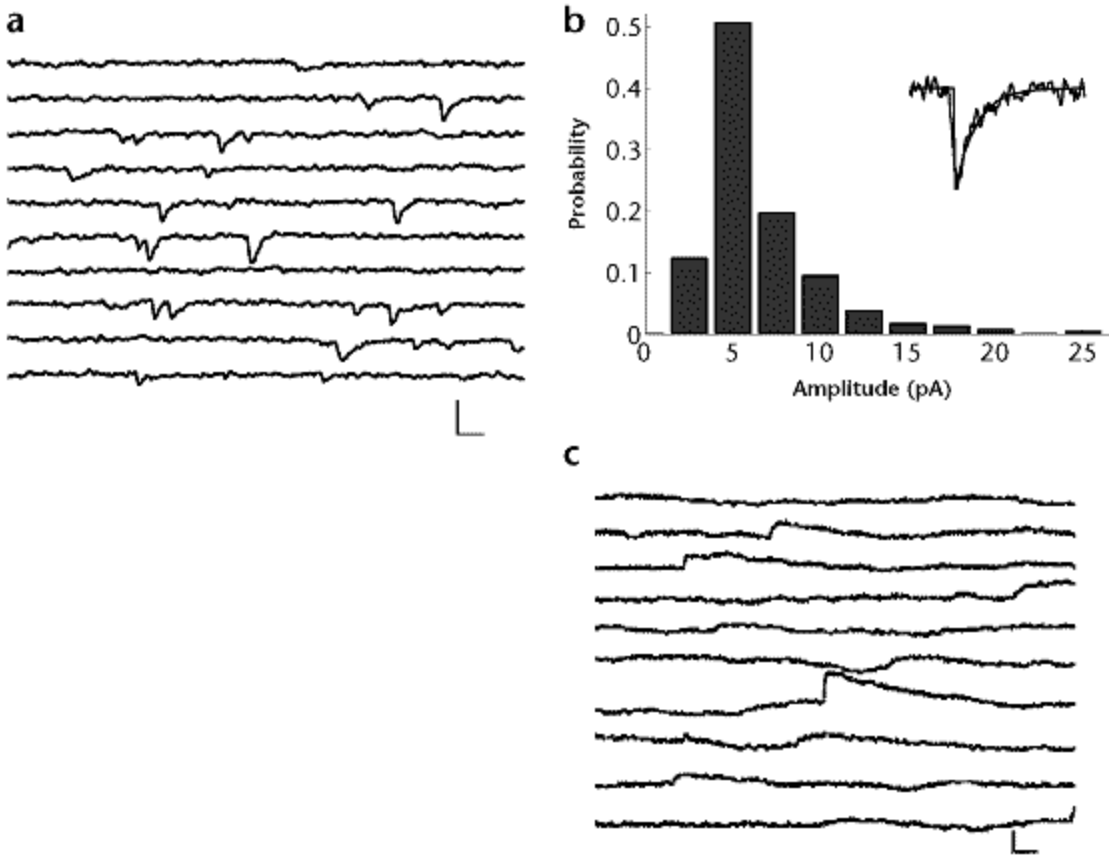


Fig. 2. Responses of a neuron from area MT of monkey extrastriate visual cortex to 210 presentations of an identical pattern of dynamic random dots. **(a)** Response rasters show time of individual spikes on each trial (row). The response varies systematically during each two second trial, producing an appearance of vertical contours throughout the raster. **(b)** Peristimulus time histogram. The instantaneous spike rate is computed in 2 ms bins. The response rate undergoes fluctuations over coarse and fine time scales. Power spectral density analysis of this waveform reveals reliable changes in instantaneous rate at up to 80–100 Hz. Although the instantaneous rate reflects changes in the visual stimulus, the exact timing of spikes is random. Each row in (a) provides a very noisy estimate of the waveform in (b). The spike trains have the characteristic of a non-homogeneous random (Poisson) point process, consistent with Fig. 1c.

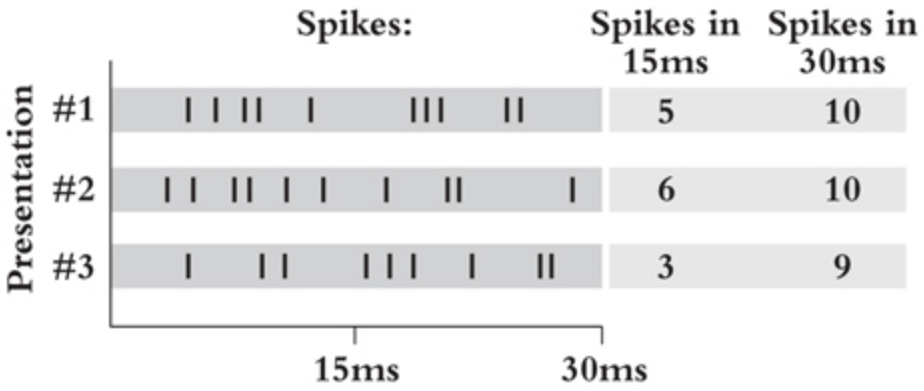


Input synchrony and the irregular firing of cortical neurons

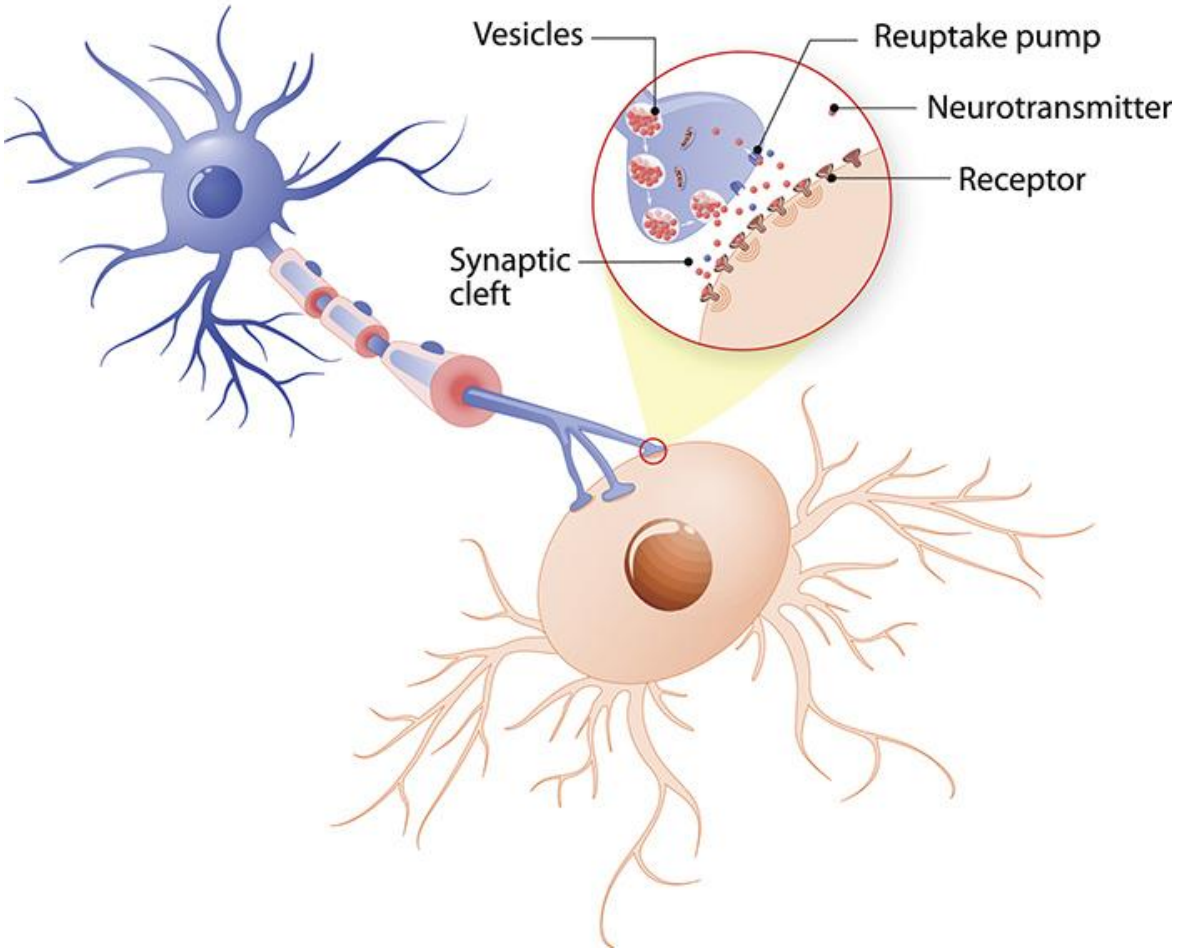
Charles F. Stevens & Anthony M. Zador

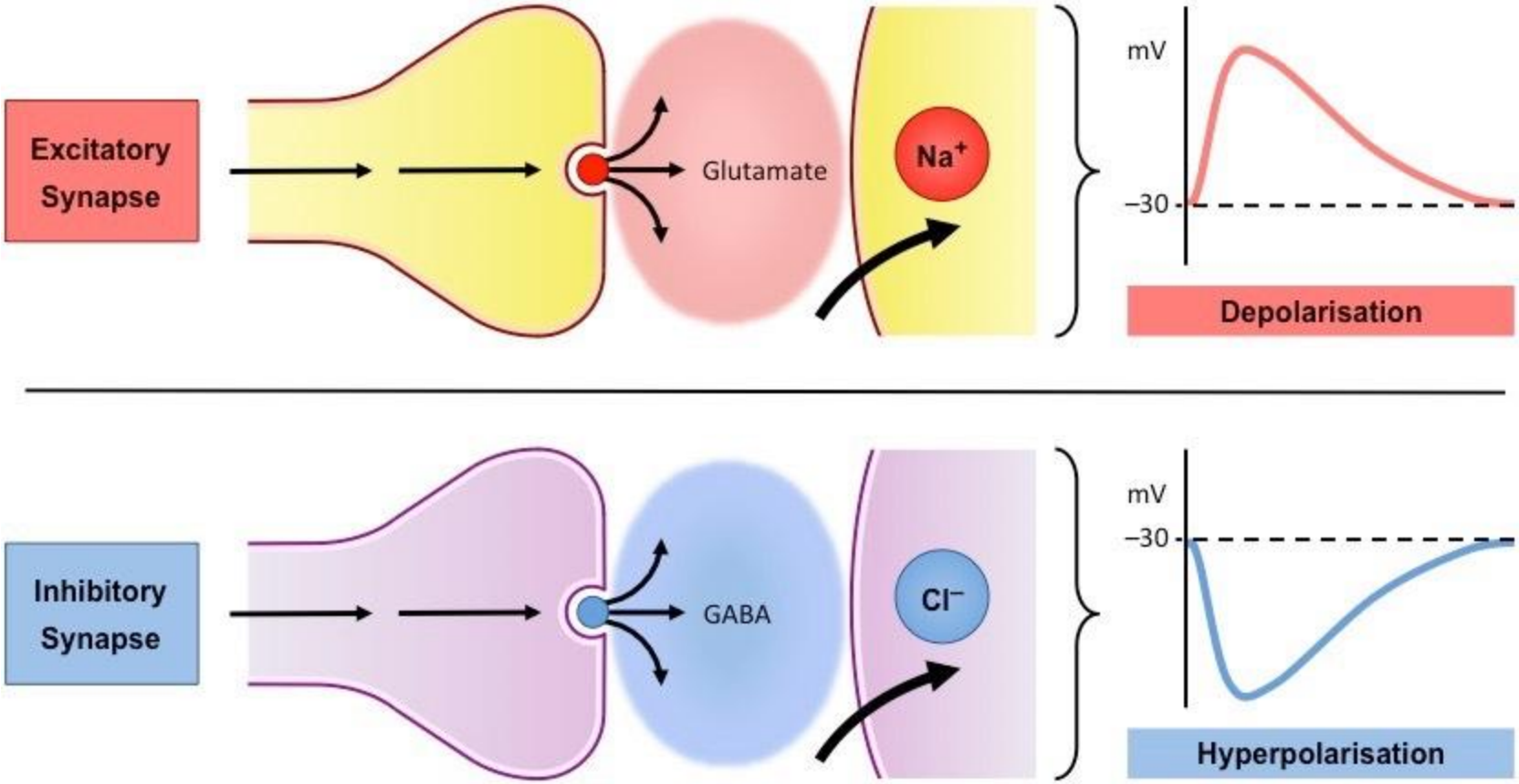
Neurons are noisy

Even though their integrative nature should remove noise.

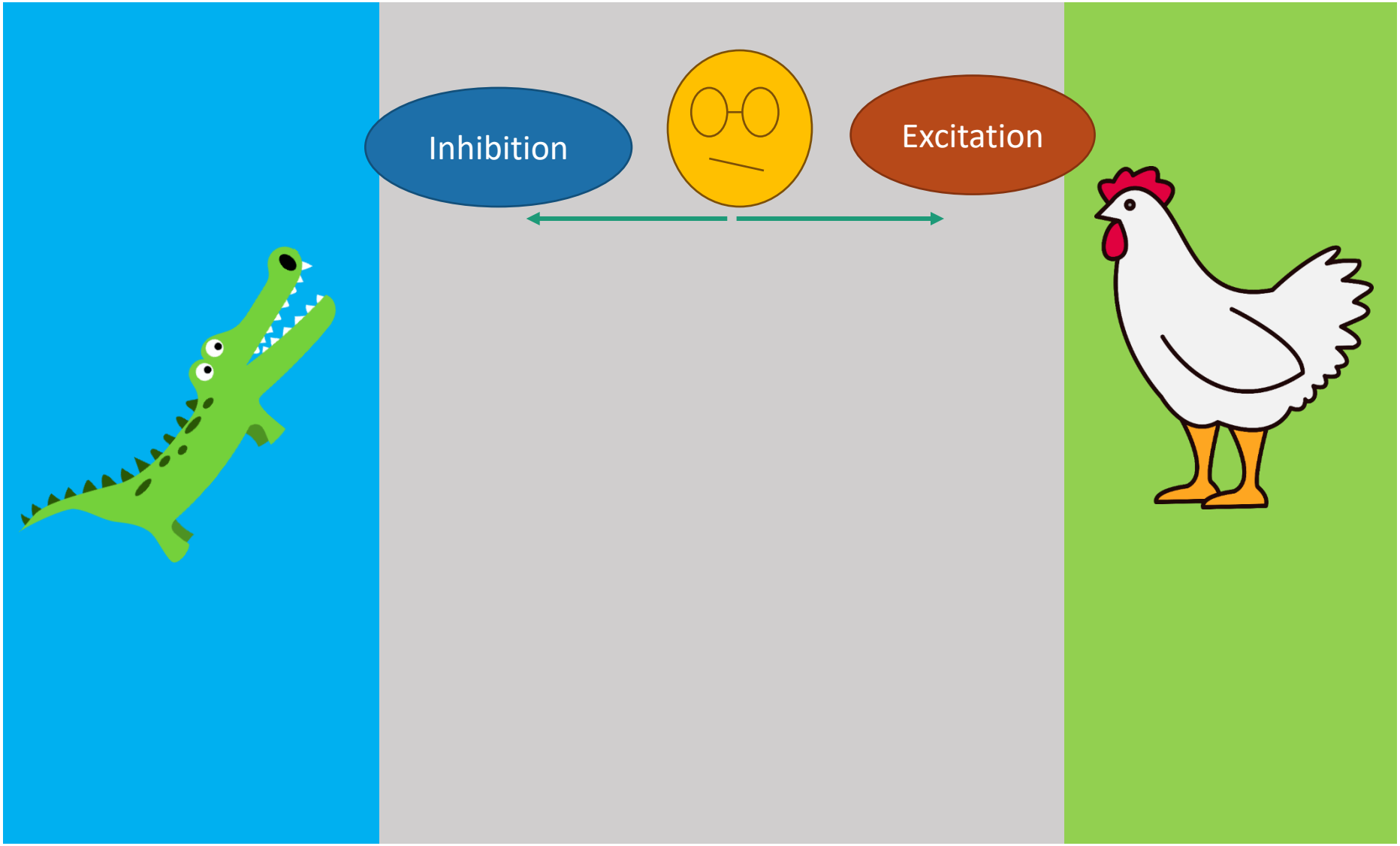


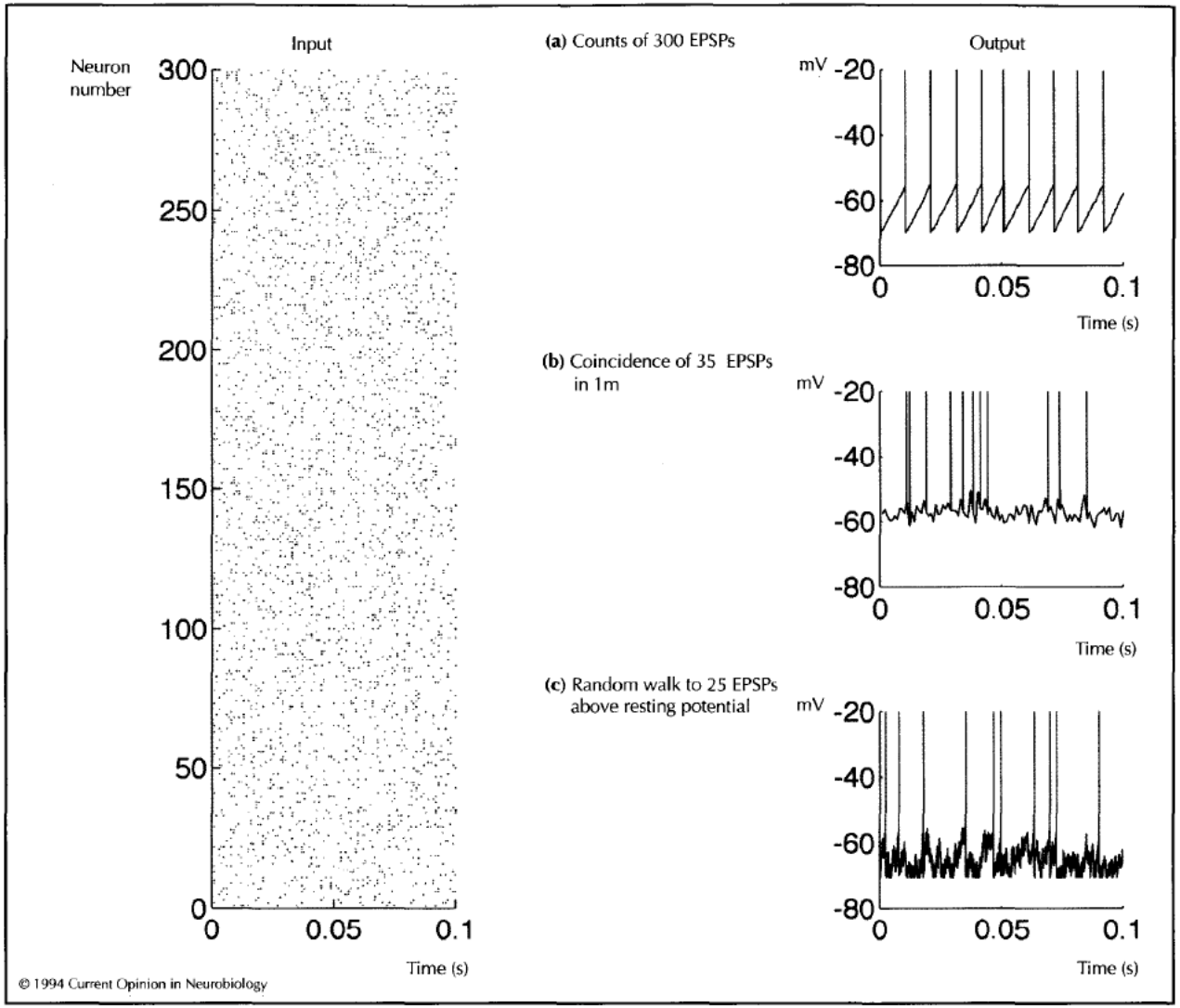
Excitation and Inhibition




Excitatory Synapse
Glutamate
Na+
mV
–30
Depolarisation
Inhibitory Synapse
GABA
Cl–
mV
–30
Hyperpolarisation




Inhibition
Excitation



Only excitation

Coincidence Detector

Excitation and InhibitionNoise, neural codes and cortical organization
Michael N.ShadlenWilliam T.Newsome 1994

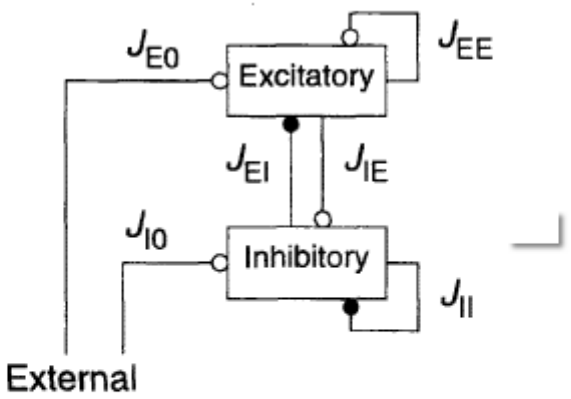
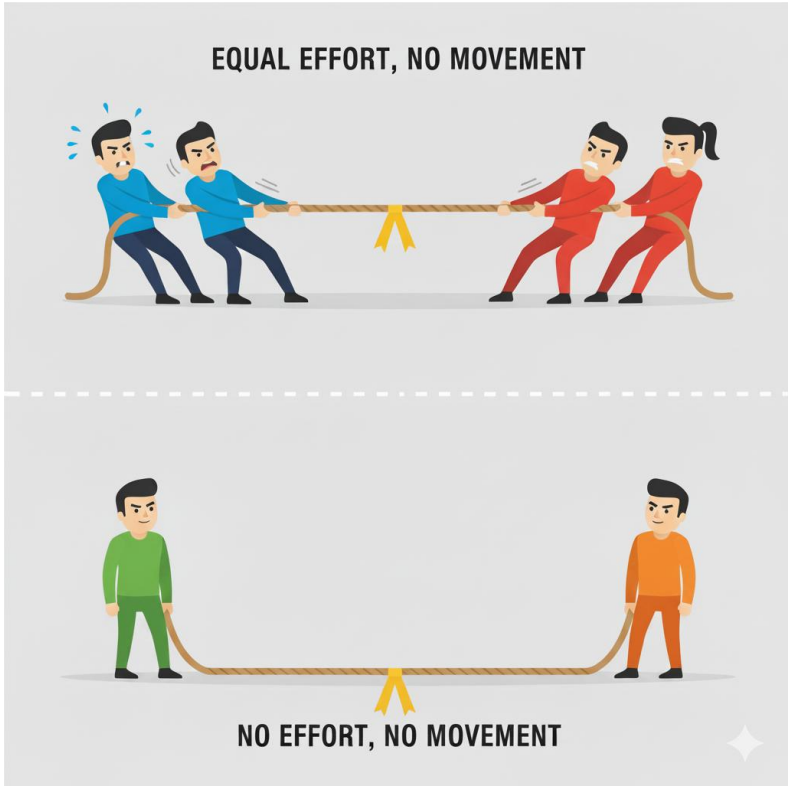


Fig. 1. Schematic representation of the network architecture. Excitatory connections between populations are shown as open circles; inhibitory ones, as filled circles.

Chaos in Neuronal Networks with Balanced Excitatory and Inhibitory Activity C. van Vreeswijk and H. Sompolinsky 1996



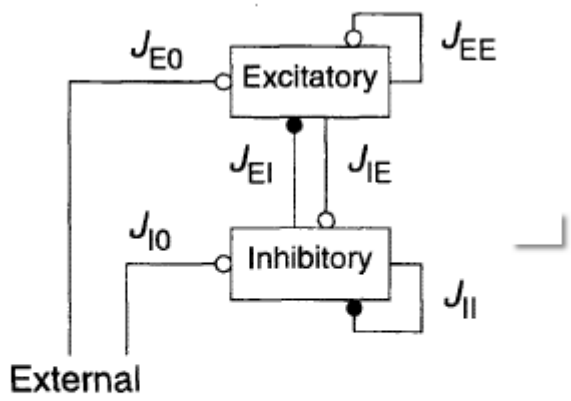
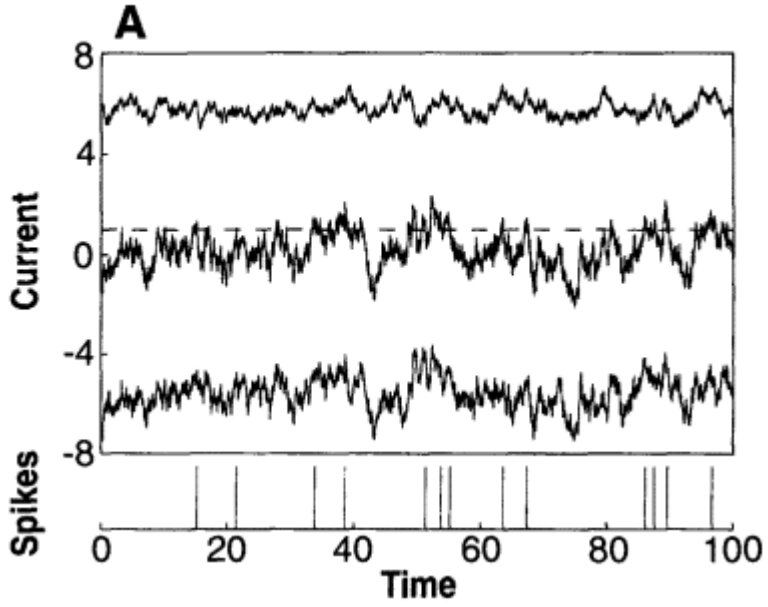
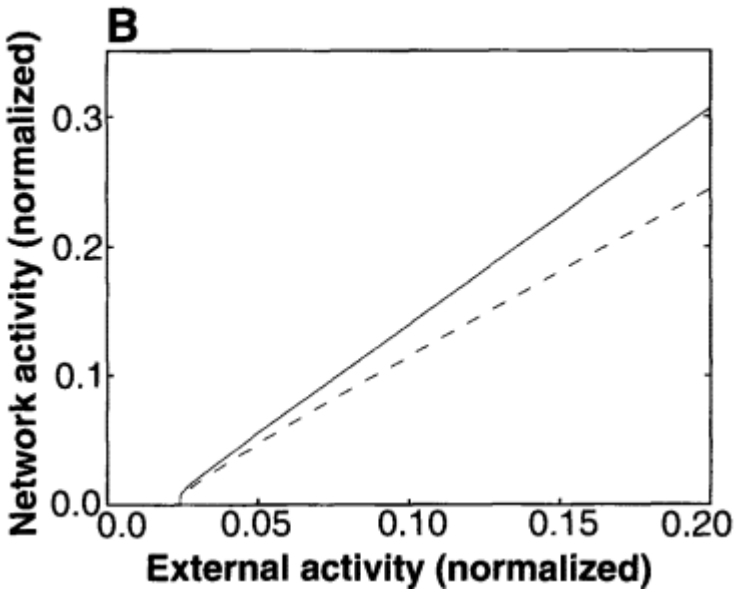


Fig. 1. Schematic representation of the network architecture. Excitatory connections between populations are shown as open circles; inhibitory ones, as filled circles.



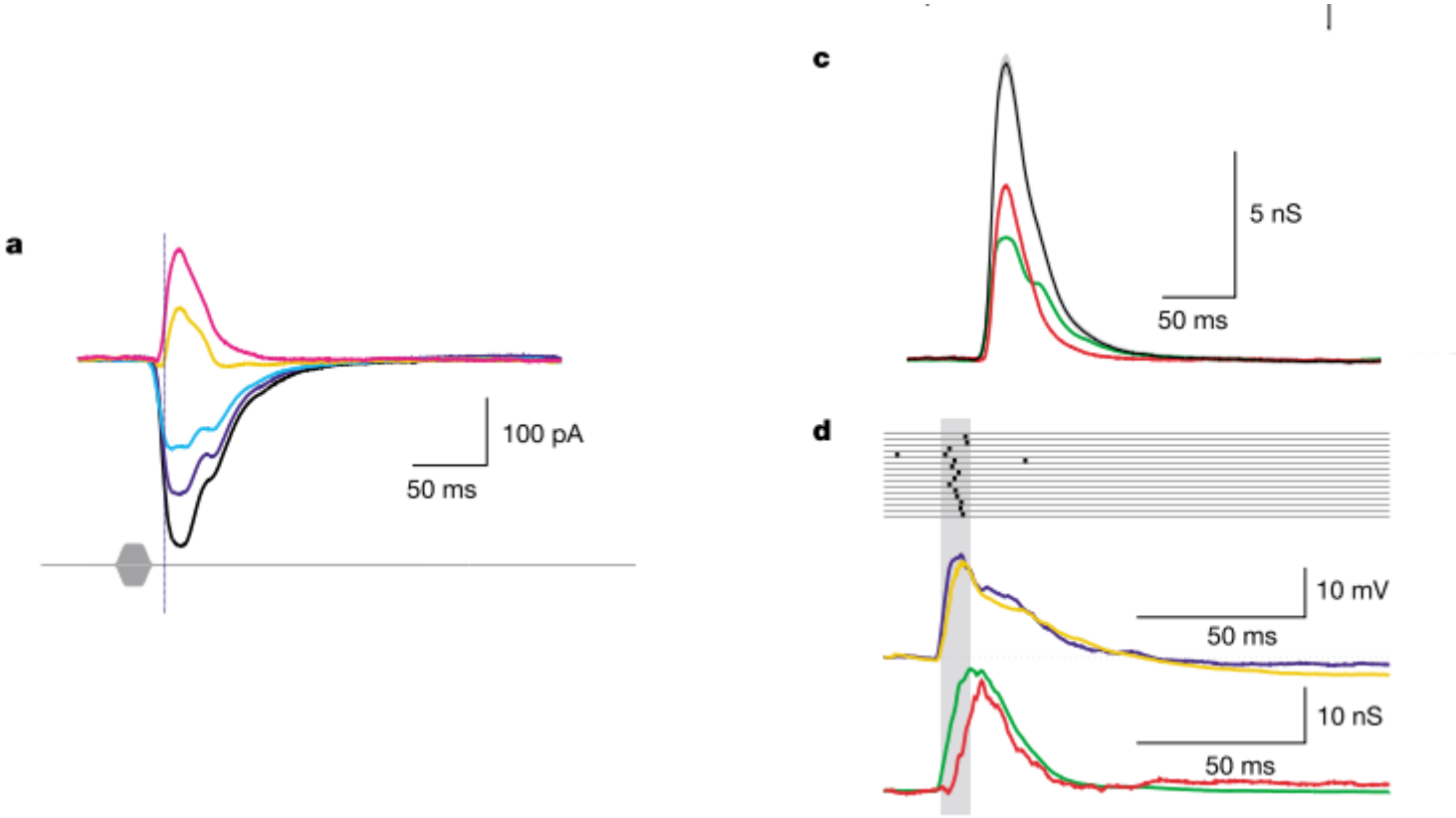


$$1 << K << N$$

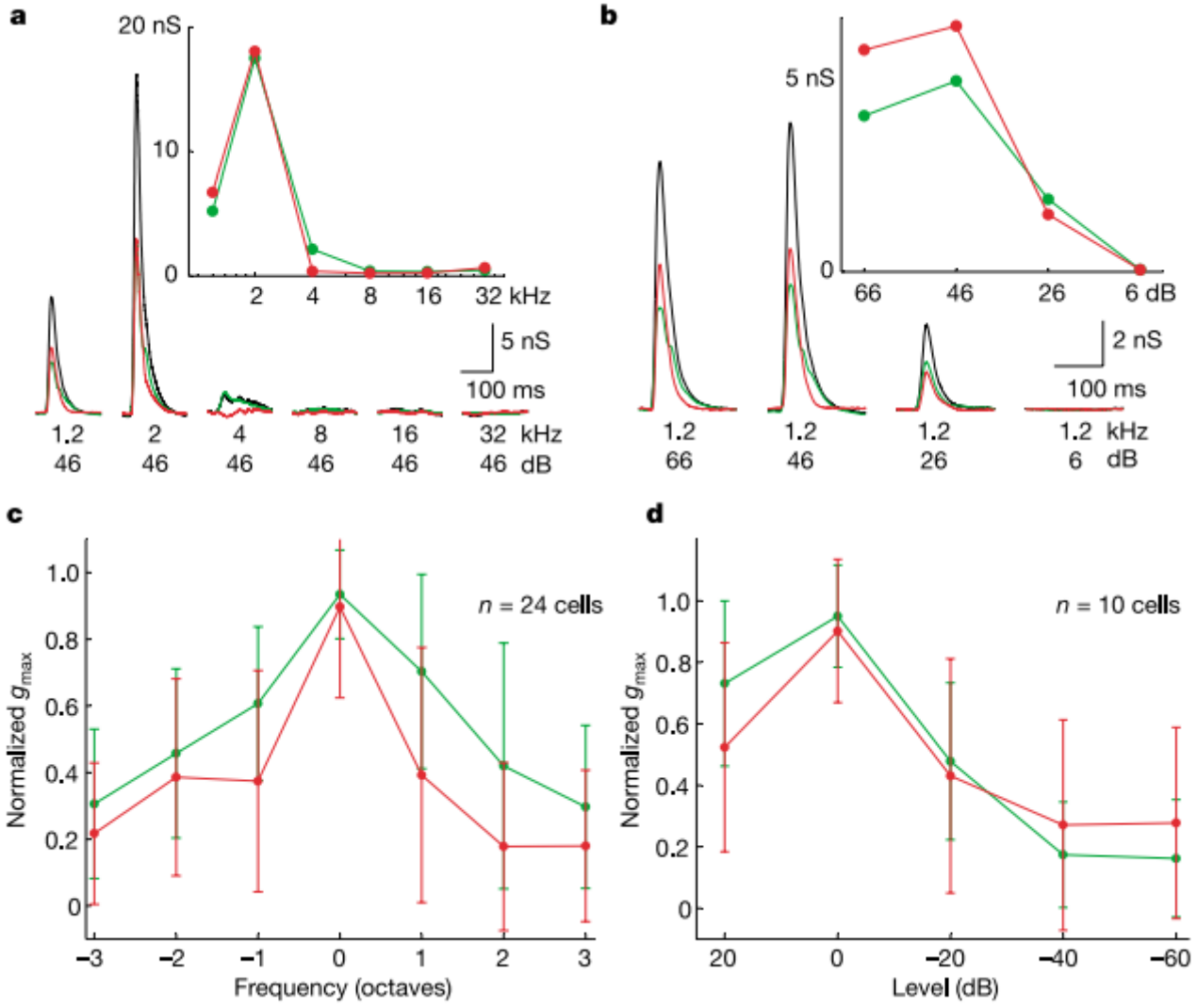
$$\mu_j = \sqrt{K}(X_j m_X + m_E - W_{jI} m_I) - \theta_j$$

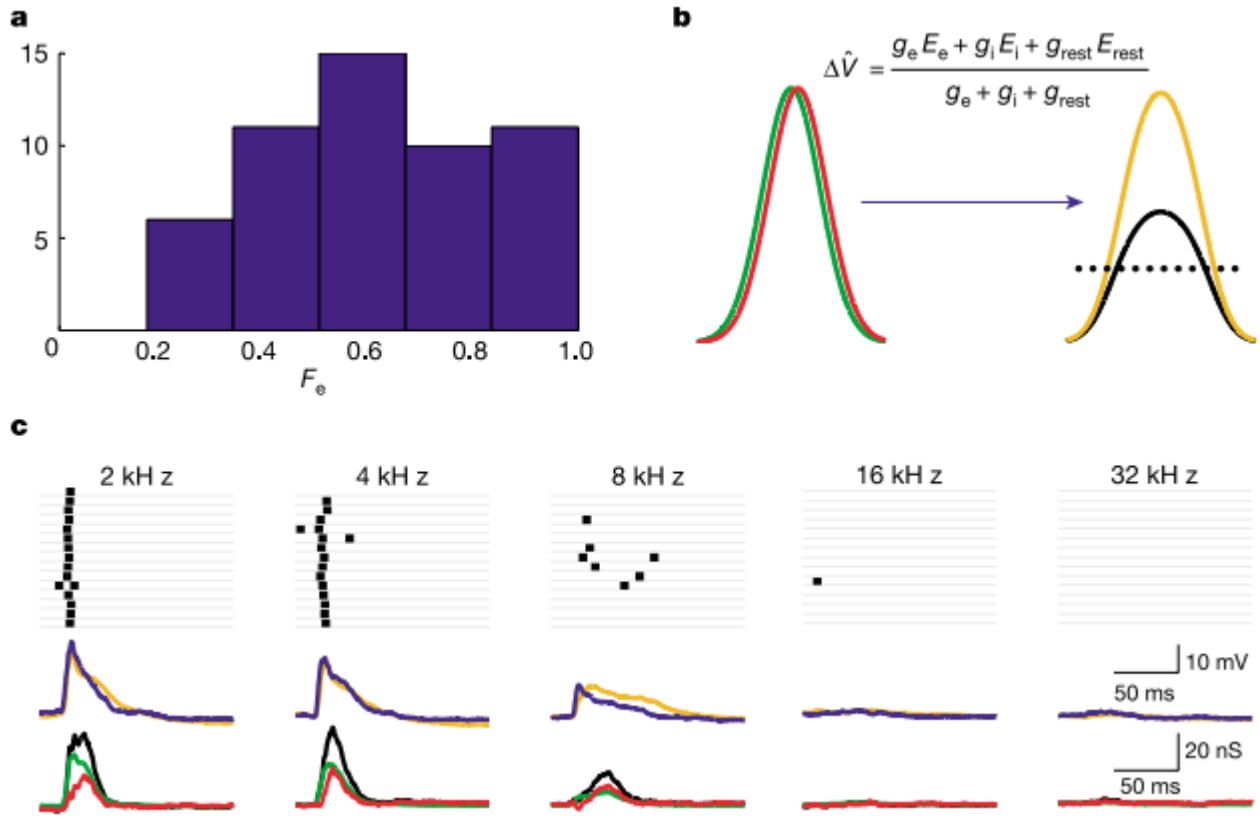
$$\sigma_j^2 = m_E + W_{jI}^2 m_I$$

$$m_j = H\left(\frac{-\mu_j}{\sqrt{\sigma_j^2}}\right)$$

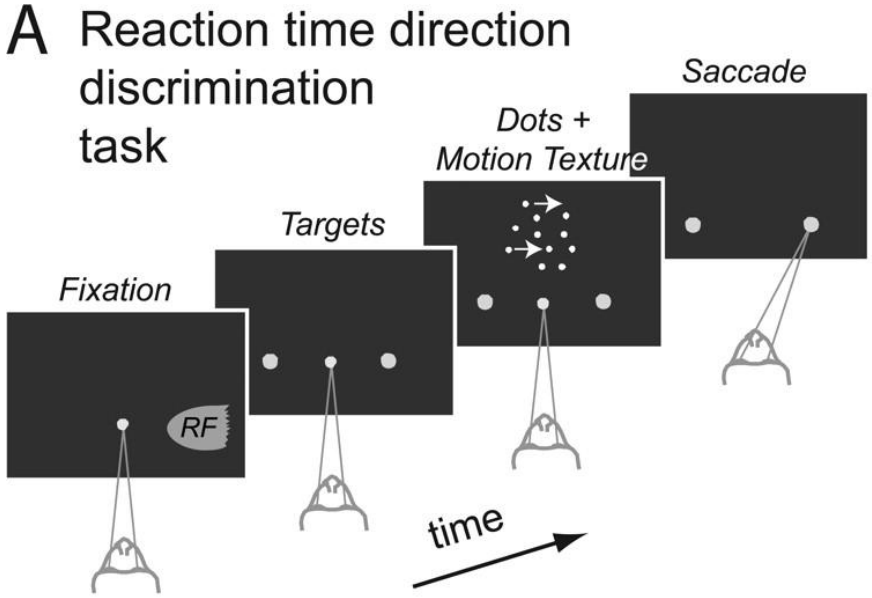


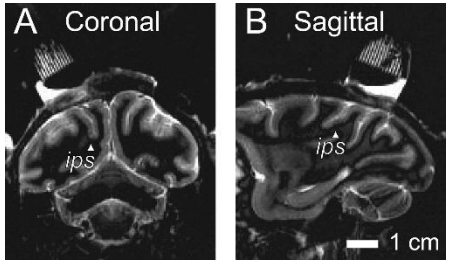
Balanced inhibition underlies tuning and sharpens spike timing in auditory cortex
Michael Wehr & Anthony M. Zador
2003


a
20 nS
10
0
2 4 8 16 32 kHz
5 nS
100 ms
1.2 2 4 8 16 32 kHz
46 46 46 46 46 46 dB
b
5 nS
0
66 46 26 6 dB
2 nS
100 ms
1.2 1.2 1.2 1.2 kHz
66 46 26 6 dB
c
n = 24 cells
Normalized g_{max}
1.0
0.8
0.6
0.4
0.2
0
−3 −2 −1 0 1 2 3
Frequency (octaves)
d
n = 10 cells
Normalized g_{max}
1.0
0.8
0.6
0.4
0.2
0
20 0 −20 −40 −60
Level (dB)


a
15
10
5
0
0.2
0.4
0.6
0.8
1.0
F_e
b
$\Delta\hat{V} = \frac{g_e E_e + g_i E_i + g_{rest} E_{rest}}{g_e + g_i + g_{rest}}$
c
2 kH z
4 kH z
8 kH z
16 kH z
32 kH z
10 mV
50 ms
20 nS
50 ms

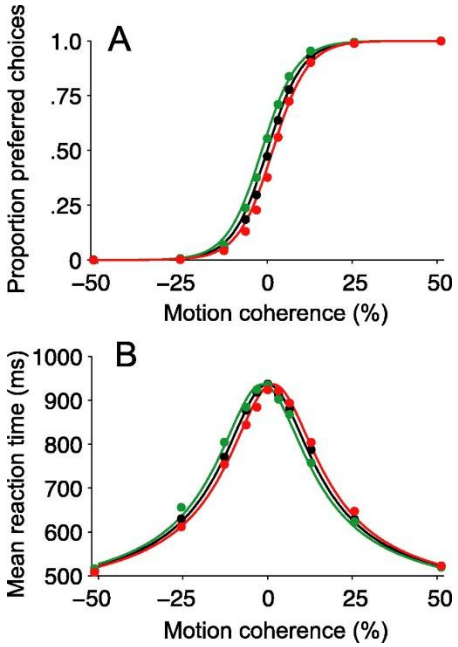
Neurons and Evidence Accumulation





Neural Activity in Macaque Parietal Cortex Reflects Temporal Integration of Visual Motion Signals during Perceptual Decision Making
Huk and Shadlen 2005

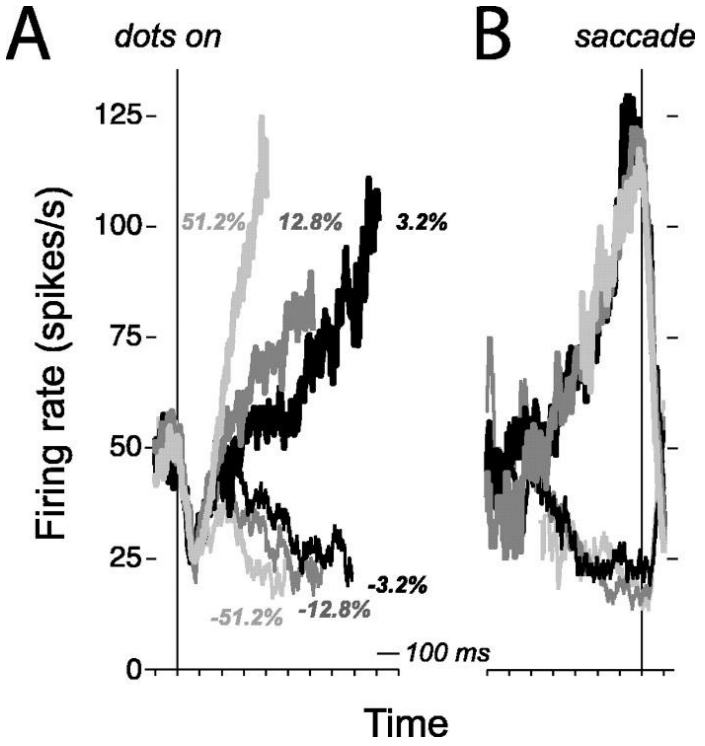
Neurons and Evidence Accumulation

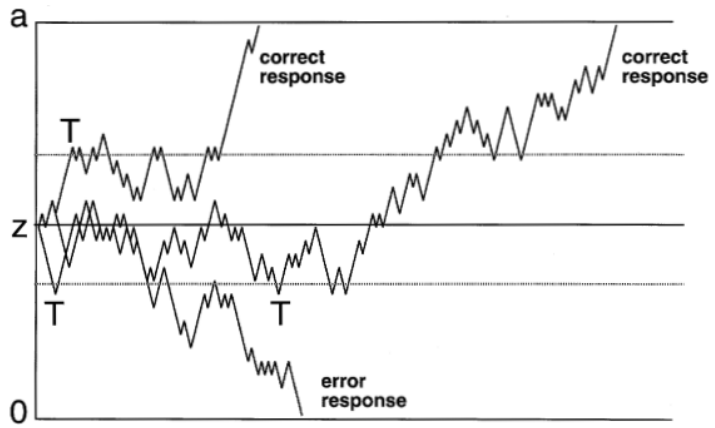


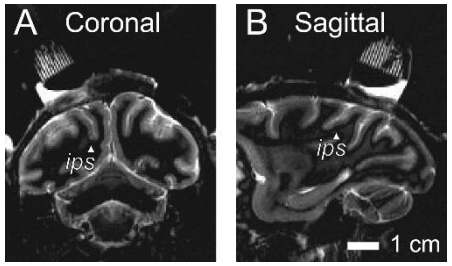
- Sensitivity of decisions

Neural Activity in Macaque Parietal Cortex Reflects Temporal Integration of Visual Motion Signals during Perceptual Decision Making
Huk and Shadlen 2005

Neurons and Evidence Accumulation

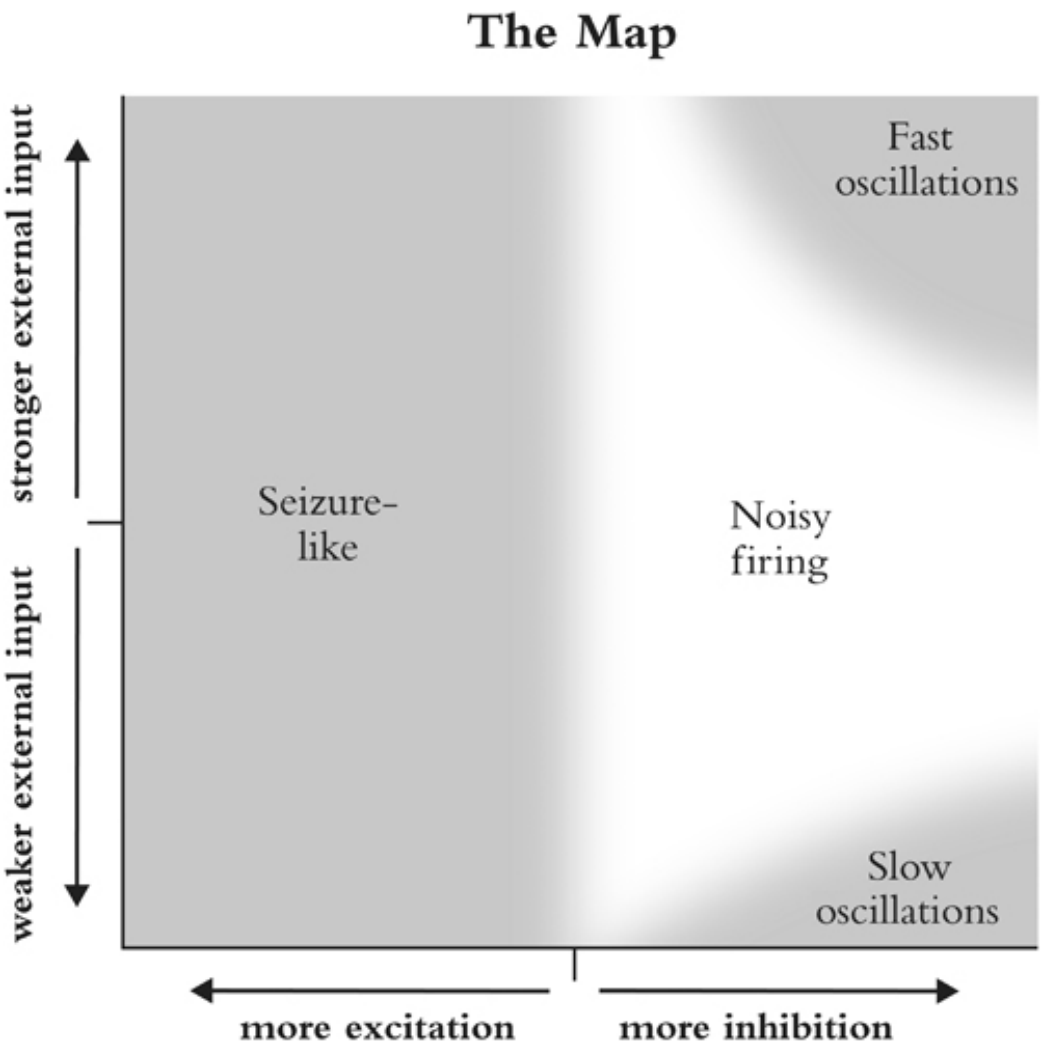




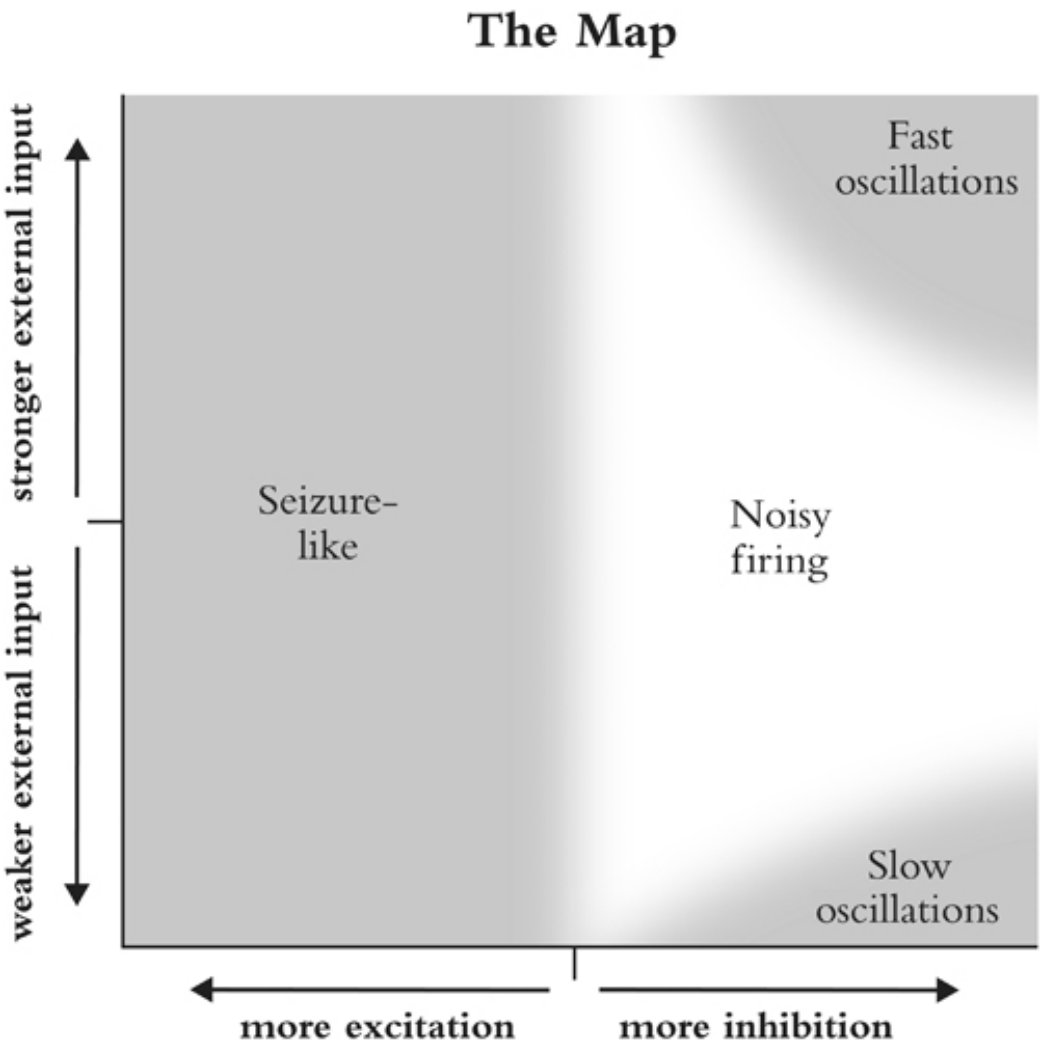


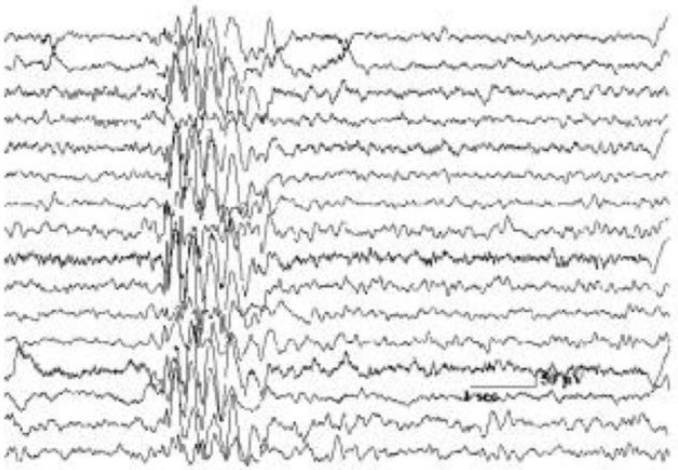
- Neurons activity in IPS (firing rates) followed the coherence of dots.
- Decision time was faster for higher coherence.
- Decision boundary was the same across coherence.

Neural Activity in Macaque Parietal Cortex Reflects Temporal Integration of Visual Motion Signals during Perceptual Decision Making
Huk and Shadlen 2005

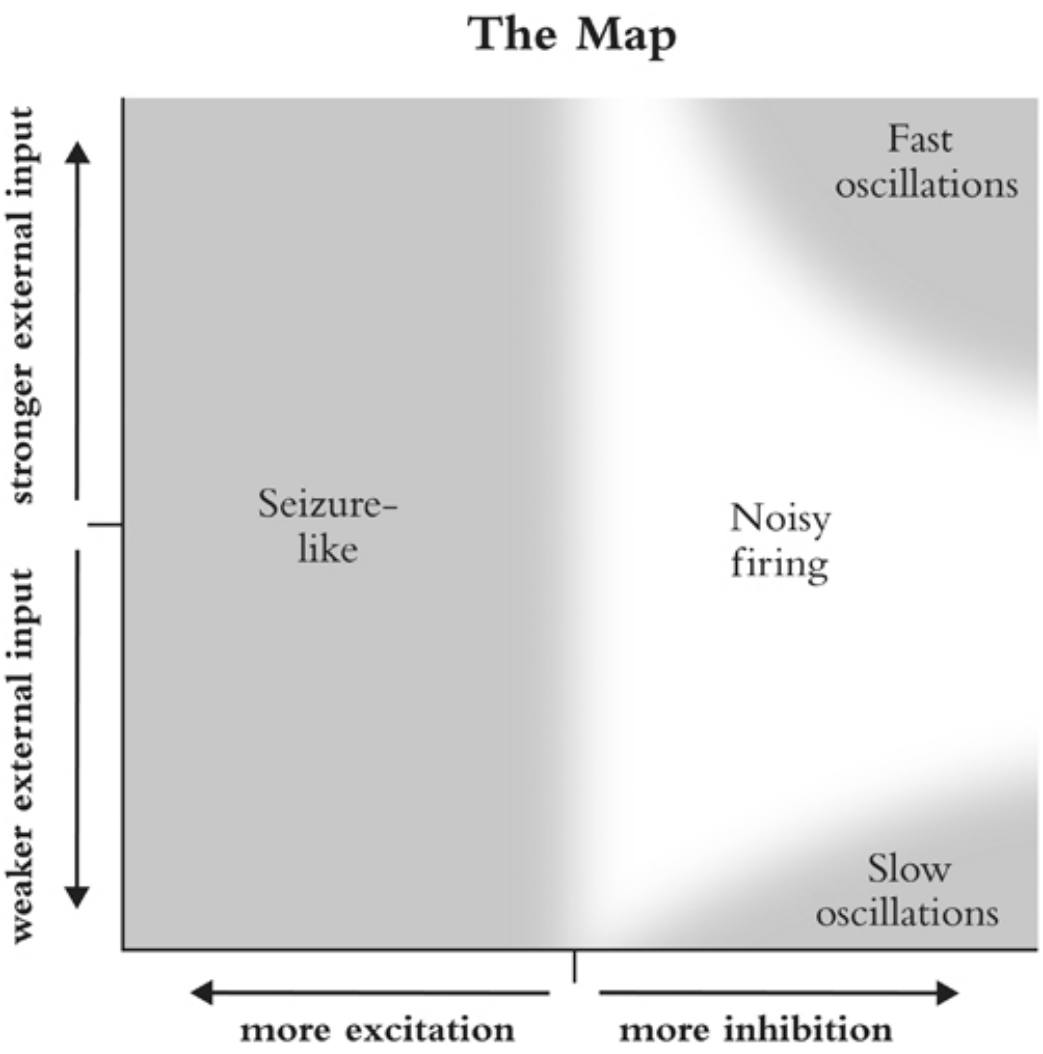

The Map
stronger external input
weaker external input
Fast oscillations
Seizure-like
Noisy firing
Slow oscillations
more excitation
more inhibition

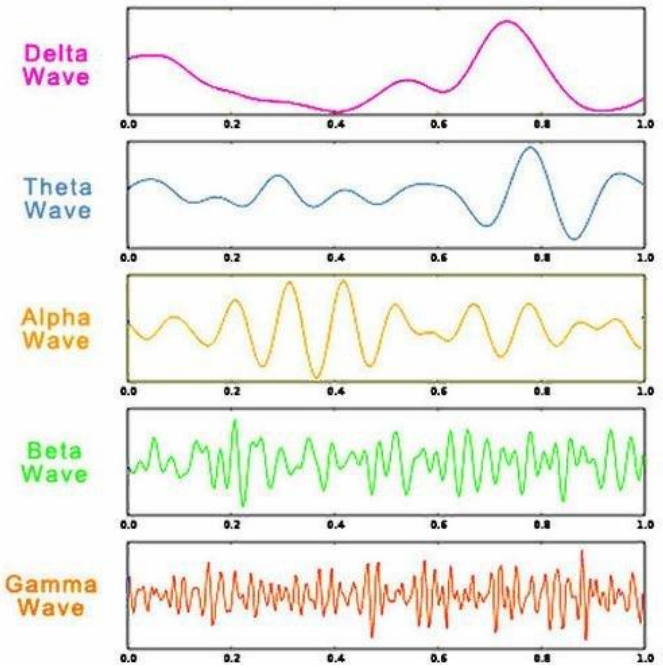
Brunel 2000


The Map
stronger external input
weaker external input
Fast oscillations
Seizure-like
Noisy firing
Slow oscillations
more excitation
more inhibition

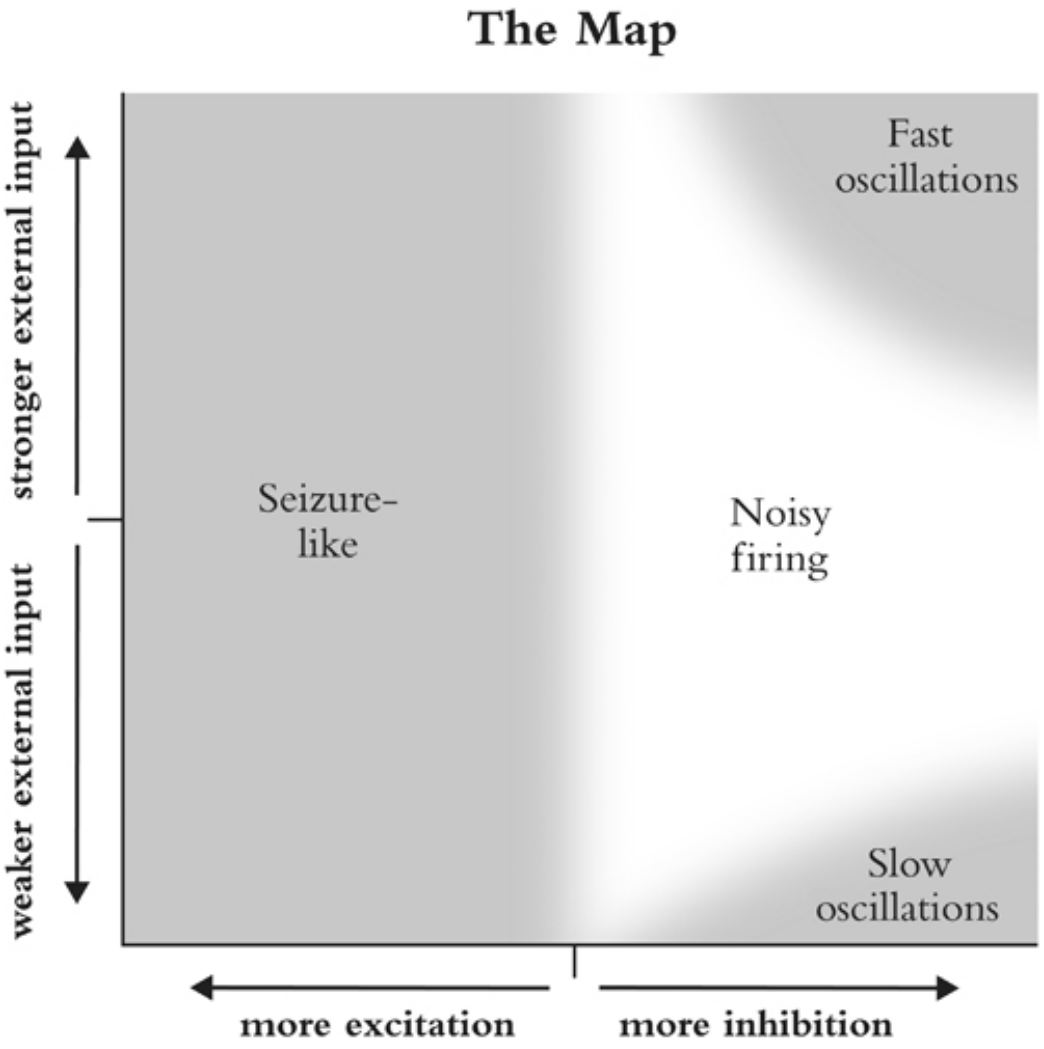


Brunel 2000


The Map
stronger external input
weaker external input
Fast oscillations
Seizure-like
Noisy firing
Slow oscillations
more excitation
more inhibition


Delta Wave
Theta Wave
Alpha Wave
Beta Wave
Gamma Wave
0.0
0.2
0.4
0.6
0.8
1.0

Brunel 2000



What is the most stable point?

Where should the system stay most of the time?